§2 Formal definition of an ACS — part 2 of 5

`Reference: subsec:self_formal_definition`

Source: `paper/section2_definition.tex`, lines 721–1578 (§2.5), at commit `e6e12cd`. The Japanese manuscript is authoritative. Part 1 is `02_definition_1.md`.

2.5 Structural self and temporal self-identity

§2.5 defines the "self" of an ACS individual formally. The self here is not subjective consciousness or identity of memory, but the structural basis on which an ACS is identified as the same individual. The self is organized at three levels: the **structural self** (the individual itself), the **dynamic self-state** (the state at each time) and the **self-environment boundary** (the distinction between self and environment). Separately from these three levels, a **realization level** is introduced, which records whether the individual is realized in well-formed fashion at each time; it supplies the vocabulary used in §2.6 to constrain why each factor of E is called "required". At the end, as different objects, the subsection defines **functional termination**, based on the activity of the basic functions the structural individual has for each primitive division and on the base functions supporting that activity, and **collapse**, based on the organizational relations that constitute the identity of the structural individual. The internal output recording whether information was discriminated as belonging to the self side or the environment side is referred to as a discrimination component of §4.

New symbols: the structural self $S_{\mathrm{struct}}(\Sigma_i)$, the dynamic self-state $S_{\mathrm{dyn},t}$, the self–environment boundary B_{SE}, the well-formed realization indicator $\mathrm{WF}_{i,t}$, the functional-termination judgment time domain $\mathbb{T}_{\mathrm{FT},i}$, the functional-termination state indicator $\mathrm{FTS}_{i,t}$, the functional-termination indicator $\mathrm{FT}_{i,t}$, the collapse-state indicator $\mathrm{CS}_{i,t}$, the collapse indicator $\mathrm{Col}_{i,t}$, and the set of admissible external actions $\Upsilon_{\mathrm{adm},i}$. The component $\xi_{\mathrm{bnd},t}$ defined in §4 is referred to as the result of cognitive boundary-attribution discrimination.

2.5.1 Structural self and well-formed realization

Definition (`def:s_struct`): structural self. The structural self of Σ_i is

$$S_{\mathrm{struct}}(\Sigma_i) := \Sigma_i,$$

the 7-tuple of `def:acs`. It is a formal structure independent of time. The index i identifies the individual and does not denote a power. The structural self gives the basis on which Σ_i is identified as the same individual. However, the existence of the 7-tuple as a mathematical object is distinguished from the structure it describes being realized in well-formed fashion at each time and continuing across times as the same ACS individual. The indicator of realization at each time is introduced in `def:wf_realization`, identity across times is organized in `def:time_identity`, and their relation to existence as a mathematical object is stated in `rem:math_vs_realization`. Σ and $S_{\mathrm{struct}}(\Sigma)$ without index abbreviate the indexed notation when only one individual is considered.

Definition (`def:wf_realization`): well-formed realization indicator (classification C). When Σ_i is an ACS in the sense of `def:acs`, for each discrete time t an indicator $\mathrm{WF}_{i,t} \in \{0,1\}$ is introduced, recording whether the structure corresponding to Σ_i is realized in well-formed fashion at t. $\mathrm{WF}_{i,t} = 1$ means that at t the seven components of Σ_i are given in the realization as their types prescribe; $\mathrm{WF}_{i,t} = 0$ means that this fails. $\mathrm{WF}_{i,t}$ is fixed separately for each time and does not by itself state whether realizations at different times belong to the same individual; identity across times is organized separately by the preliminary conditions of `def:time_identity`. $\mathrm{WF}_{i,t}$ is not a component of the 7-tuple but a time-dependent indicator describing the success or failure of realization. Phase 1 gives no operational procedure for judging its value from finite observation; that is a task for Phase 2.

Mathematical level versus realization level (`rem:math_vs_realization`). $S_{\text{struct}}(\Sigma_i) = \Sigma_i$ is a time-independent mathematical object, and its existence does not depend on the value of $\mathrm{WF}_{i,t}$. When $\mathrm{WF}_{i,t} = 0$, the 7-tuple does not vanish as a mathematical object; what is lost is the state in which the structure it describes continues to be realized as the same ACS individual. "Persistence" in the paper means not that Σ_i continues to exist as a mathematical object, but that $\mathrm{WF}_{i,t} = 1$ is kept at each time and the preliminary conditions of temporal self-identity (`def:time_identity`) continue to hold. This corresponds to the different roles of constitutive requirements and realization indicators: CR-1 and CR-2 are not diagnostic values at individual times but conditions placing the 7-tuple in the class of ACSs; $\mathrm{WF}_{i,t}$ records whether the realization of an ACS individual so determined holds at time t. Accordingly, CR-1's requirement that each factor of E be "necessary for the persistence of the ACS" is stated not as a condition on the existence of the mathematical structure Σ_i containing E, but as a condition on the course of $\mathrm{WF}_{i,t}$. With this separation of levels, no circularity arises in which the semantic constraint on E presupposes the existence of an object containing E.

2.5.2 Dynamic self and the self–environment boundary

Definition (`def:s_dyn`): dynamic self-state. The dynamic self-state of Σ_i at time t is

$$S_{\text{dyn},t} := (x_t,\ A_t),$$

where $x_t \in X$ is the state vector and A_t the feasible action set (`def:at`). $A_t = \emptyset$ does not mean that $S_{\text{dyn},t}$ vanishes; it means that no feasible action exists (blockage, `def:blockage_sec2`), within the present framework.

Definition (`def:b_se`): self–environment boundary. Under TSC, the structural separation representing the self–environment boundary is

$$B_{SE} := (X, E_{\text{ext}}), \qquad X \cap E_{\text{ext}} = \emptyset,$$

and for realized states $x_t \in X$, $e_t \in E_{\text{ext}}$ at every t. In Phase 1, B_{SE} is a fixed structural separation, distinguishing the type X of internal states from the type E_{ext} of external environment states in the set representation. It does not claim a physical or spatial boundary, or an ontologically complete division between self and non-self. The required environment space E is a different object from the type separation represented by B_{SE}: B_{SE} distinguishes the internal-state type from the external-state type, while E, as a set of satisfaction maps on pairs of external and internal states, represents the ACS's dependence for persistence on the external world. E is therefore not a subset of E_{ext} and is not identified with the second component of B_{SE}.

Reference to boundary-attribution discrimination (`rem:xi_bnd_reference`, classification B). The cognitive result of discriminating self from environment is defined in §4 as the boundary-attribution component of the overall row $\xi_t(*)$ of M1-b's discrimination state ξ_t:

$$\xi_{\text{bnd},t} \in \Xi_{\text{bnd}} = \{\mathsf{self}, \mathsf{env}, \mathsf{und}\}.$$

§2 refers to it without redefining it. self means that the information concerned was discriminated as belonging to the self side, env to the environment side, and und that neither was settled. The value of $\xi_{\text{bnd},t}$ represents the content of discrimination, not the correct operation of the discrimination function. The operation level of M1-b and the health of its implementation parts are described separately by the functional axes of §4 and the health indices of §5.

Proposition (`prop:xi_se_independence`): independence of $\xi_{\text{bnd},t}$ and S_{struct}. Under TSC, the existence of the structural self $S_{\text{struct}}(\Sigma_i)$ does not depend on the value of $\xi_{\text{bnd},t}$ (within the present framework). *Proof sketch:* $S_{\text{struct}}(\Sigma_i) = \Sigma_i$ is a time-independent mathematical object, and $\xi_{\text{bnd},t}$, an output of M1-b at time t, does not appear in its definition; so no value of $\xi_{\text{bnd},t}$ has logical implications for its existence.

Failure of cognitive discrimination and independence of structure (`rem:xi_se_interpretation`). If M1-b's discrimination does not operate correctly, a state may arise in which the ACS cannot reliably use the discrimination of self and environment. §2 does not state as a general consequence the effect of such a state on the maintenance of feasibility, blockage or well-formed realization, and the paper does not describe it in Phase 1. Since K and the

feasible action set are fixed independently of the internal-model state (`def:at`), the result of discrimination bears on the feasibility of the state only through the selection of the effective action (§4). If M1-b itself is rewritten from outside, a state misclassifying the structural separation B_{SE} can arise in theory, whatever the output value. In Phase 1 such an inconsistency between B_{SE} and M1-b's classification is recorded as an intermediate state without an operational criterion; introducing a mechanism for checking the consistency of M is a task for Phase 2. `prop:xi_se_independence` concerns only the existence of $S_{\mathrm{struct}}(\Sigma_i)$; it does not state that the value of $\mathrm{WF}_{i,t}$ is independent of $\xi_{\mathrm{bnd},t}$.

2.5.3 Non-branching of successor realizations

Assumption (`ass:no_branching`): non-branching of the individual-identity relation. In tracking the temporal self-identity of Σ_i, for an ACS individual realization at time t there is at most one ACS individual realization at time $t+1$ that is matched to it as the continuation of the same individual.

This assumption is separate from the single-valuedness of $f : X \times U \times E_{\mathrm{ext}} \to X$. Single-valuedness fixes the next-state value x_{t+1} uniquely for a fixed input (x_t, u_t, e_t), but it does not exclude several individual realizations with the same next-state value, or the creation of copy individuals.

Introduction status: A (axiomatic). Strength: strong. It excludes from Phase 1 the case in which successor realizations matched as the continuation of the same ACS individual branch into several. It does not deny mathematically that copy individuals can be created; it means that Phase 1 does not define individual identity when a copy or several successor realizations arise. Allowing branching of identity requires reformulating the temporal self-identity relation, the structural self and the correspondence of successor individuals; this is treated as an open problem in §9.

2.5.4 Preliminary conditions of temporal self-identity

Definition (`def:time_identity`): preliminary conditions of temporal self-identity. This definition is a preliminary organization anticipating the definition of collapse (`def:collapse`) and the introduction of the time horizon $\hat{\tau}_t$ in §5. §2 does not claim full existence, uniqueness or reachability of temporal self-identity. The preliminary conditions for the dynamic self-states of Σ_i at t and $t+1$ to **belong to the same ACS individual** are that the following three conditions hold together (under stated assumptions).

- **(TI-1) Necessary condition of structural continuity.** As a necessary condition for the individual realization carrying the structural self to continue as the same individual from t to $t+1$, it is required that $\mathrm{WF}_{i,t} = 1$ and $\mathrm{WF}_{i,t+1} = 1$. This states that well-formed realization holds at both times. Continuation as the same individual is not represented by WF alone; it is organized as the preliminary conditions (TI-1)–(TI-3) together. The distinction from functional termination, collapse and blockage is stated in `rem:time_identity_collapse`.
- **(TI-2) Transition consistency of the dynamic self-state.** The continuation from $S_{\mathrm{dyn},t} = (x_t, A_t)$ to $S_{\mathrm{dyn},t+1} = (x_{t+1}, A_{t+1})$ is required to be fixed by the same ACS dynamics: the next state is generated by $x_{t+1} = f(x_t, u_t, e_t)$, and, under the realized external environment state $e_{t+1} \in E_{\mathrm{ext}}$ at $t+1$, the feasible action set is fixed as

  $$A_{t+1} = \{u \in U \mid f(x_{t+1}, u, e_{t+1}) \in K\}.$$

  So $(x_t, A_t) \to (x_{t+1}, A_{t+1})$ is described on the same state transition system, consistent with f and the definition of the feasible action set (`def:state_transition`, `def:at`). This condition does not require A_{t+1} to be nonempty. $A_{t+1} = \emptyset$ only means blockage at $t+1$ (`def:blockage_sec2`); the paper does not state as a general consequence from this the vanishing of $S_{\mathrm{dyn},t+1} = (x_{t+1}, \emptyset)$ itself, or $\mathrm{WF}_{i,t+1} = 0$.
- **(TI-3) Non-branching of the individual-identity relation.** For an ACS individual realization at time t, at most one realization at $t+1$ is matched to it as the continuation of the same individual (`ass:no_branching`). Separately from the determinism of f, this is an assumption on individual identity that excludes from Phase 1 the case where several successor individuals arise from the same ACS individual.

Branching that produces copy selves is excluded from Phase 1 explicitly by `ass:no_branching`, not by the single-valuedness of f. Relaxing non-branching and reformulating temporal self-identity to allow branching are treated as the open problem `prob:branching_identity` in §9. In the paper, the self is defined as $S_{\text{struct}}(\Sigma_i)$, and cognitive content (the memory content of M_1) does not affect the judgment of structural persistence. Subjective self, consciousness and identity of memory are not objects of description in Phase 1 (under stated assumptions).

2.5.5 Functional termination and collapse

`Reference: subsubsec:collapse_sec2`

The termination of an ACS individual is defined in two stages.

- **Functional termination**: the state in which the activity of the basic functions the structural individual has for each primitive division (sensing, discrimination, evaluation, action), and all the base functions supporting that activity, have stopped, and none of them is established again as the same individual.
- **Collapse**: the state in which the organizational relations that had constituted the identity of the structural individual are lost in such a way that they are not recovered as the same individual. In the paper, "collapse" without qualification refers to this.

Both are state conditions for judging that an individual at a given time is already in that state, not causes or conditions of reaching termination. §2 defines them but does not claim reaching functional termination or collapse, sufficient conditions for it, or reaching times. The relation to blockage and descriptive state bands, external recoverability, and the organization of paths of reaching them are treated in §7. Collapse is defined first because the time domain on which the functional-termination state is judged is fixed using the collapse-state indicator; this does not represent a temporal order in which the two states hold.

The non-recoverability contained in functional termination and collapse is not impossibility against every conceivable physical action; it is fixed relative to the range of external actions the framework takes into account. For each Σ_i, the set of external actions considered in evaluating external recoverability is written $\Upsilon_{\text{adm},i}$, with elements $\upsilon \in \Upsilon_{\text{adm},i}$. §2 declares only the symbol and its type. $\Upsilon_{\text{adm},i}$ is a set; its topology, measure or algebraic structure is not fixed, and it is not assumed to be nonempty. Its content is fixed in §7 (`def:admissible_external_action`). υ is a variable with a different role from the $u \in U$ that the ACS selects from its action space, and it is not substituted implicitly into the arguments of f. The type of the transition map g_O of the action-response component, which appears in (SC-2) and (FT-2), is fixed in `def:env_realization`. The symbol $\Upsilon_{\text{adm},i}$ is unrelated to the range Ξ of M1-b's discrimination state in §4.

Definition (`def:collapse`): collapse-state indicator and collapse indicator. For an ACS individual Σ_i and $t \in \mathbb{N}_0$, set $\mathrm{CS}_{i,t} = 1$ when (SC-1) and (SC-2) both hold, and $\mathrm{CS}_{i,t} = 0$ otherwise. $\mathrm{CS}_{i,t} \in \{0, 1\}$ is the **collapse-state indicator**.

- **(SC-1) Loss of the organizational relations constituting structural identity.** At time t, the organizational and binding relations among the constituents of Σ_i that had constituted the identity of Σ_i as a structural individual are lost. Whether the constituents themselves remain is not at issue.
- **(SC-2) Non-recoverability as the same individual, relative to the framework.** For times from t on, within the range of the continuation by Σ_i's own ordinary autonomous update and the transition g_O of the action-response component under the realized sequence of exogenous background components $(b_s)_{s \geq t}$, and of other continuations obtained by adding admissible external actions $\upsilon \in \Upsilon_{\text{adm},i}$ to it, the framework admits no continuation in which the relations of (SC-1) are established again as the same individual Σ_i.

(SC-1) and (SC-2) are judged by a judgment rule fixed for each system before the judgment. The rule includes the criterion of individual identity in (SC-2) and is not changed afterwards according to the course or result of what is judged. Phase 1 gives no general form of this rule.

Further, for times from a time t with $\mathrm{CS}_{i,t} = 1$ on, the framework admits no realization belonging to Σ_i as the same individual (the **attribution provision**). Hence at each such time s, no functional activity or base function belonging to Σ_i holds; and since no realization of Σ_i in

the sense of `def:wf_realization` exists at s, $\mathrm{WF}_{i,s} = 0$.

The **collapse indicator** $\mathrm{Col}_{i,t} \in \{0,1\}$ of Σ_i is

$$\mathrm{Col}_{i,t} = 1 \quad :\Longleftrightarrow \quad \exists t' \in \{0,1,\dots,t\} :\ \mathrm{CS}_{i,t'} = 1,$$

and $\mathrm{Col}_{i,t} = 0$ otherwise. $\mathrm{Col}_{i,t} = 1$ is a history recording that a collapse state has held by time t. When $\mathrm{Col}_{i,t} = 1$, one says that the **collapse** of Σ_i has occurred by time t. This does not mean that at time t a realization belonging to Σ_i exists and is in a collapsed state.

Definition (`def:functional_termination`): functional-termination state indicator and functional-termination indicator. For Σ_i, the **functional-termination judgment time domain** is

$$\mathbb{T}_{\mathrm{FT},i} := \{t \in \mathbb{N}_0 \mid \forall s \in \mathbb{N}_0\ (s < t \Rightarrow \mathrm{CS}_{i,s} = 0)\}.$$

It consists of the times at which no collapse state has yet held, together with the time at which a collapse state first holds. For each $t \in \mathbb{T}_{\mathrm{FT},i}$, set $\mathrm{FTS}_{i,t} = 1$ when (FT-1) and (FT-2) both hold, and $\mathrm{FTS}_{i,t} = 0$ otherwise; $\mathrm{FTS}_{i,t} \in \{0,1\}$ is the **functional-termination state indicator**. For $t \notin \mathbb{T}_{\mathrm{FT},i}$, $\mathrm{FTS}_{i,t}$ is not defined.

- **(FT-1) Total functional stoppage.** At time t, both of the following hold.
  - (a) For the basic functions (`def:basic_functions`) that Σ_i has for each primitive division (sensing, discrimination, evaluation, action; `def:primitive_functions`), their functional activity has all stopped. This is a state condition referring to the total absence of functional activity at time t; it does not require that activity held earlier. It is not waiting or rest, such as the selection of selective waiting $u_\emptyset$ (`rem:primitive_activity_cessation`).
  - (b) The base functions (`def:base_functions`) supporting the activity of the basic functions in (a) have all stopped.
- **(FT-2) Non-recoverability as the same individual, relative to the framework.** For times from t on, within the range of the continuation by Σ_i's own ordinary autonomous update and the transition g_O under the realized sequence $(b_s)_{s \geq t}$, and of other continuations obtained by adding admissible external actions $\upsilon \in \Upsilon_{\mathrm{adm},i}$, the framework admits no continuation in which the functional activity of any basic function of (a), or any base function of (b), is established again as the same individual Σ_i.

The **functional-termination indicator** $\mathrm{FT}_{i,t} \in \{0,1\}$ is defined for every $t \in \mathbb{N}_0$ by

$$\mathrm{FT}_{i,t} = 1 \quad :\Longleftrightarrow \quad \exists t' \in \mathbb{T}_{\mathrm{FT},i} \cap \{0,1,\dots,t\} :\ \mathrm{FTS}_{i,t'} = 1,$$

and $\mathrm{FT}_{i,t} = 0$ otherwise. $\mathrm{FT}_{i,t} = 1$ is a history recording that a functional-termination state has held by time t; it does not mean that Σ_i is in a functional-termination state at t. When $\mathrm{FT}_{i,t} = 1$ and $\mathrm{Col}_{i,t} = 0$, Σ_i is said to be **after functional termination** at time t.

The judgments of functional activity and base functions in (FT-1), and the judgment of (FT-2), follow a judgment rule fixed for each system before the judgment. The rule includes the criterion of individual identity in (FT-2), is fixed consistently with the attribution provision of `def:collapse`, and is not changed afterwards according to the course or result of what is judged. Phase 1 gives no general form of this rule.

The continuations quantified in (SC-2) and (FT-2) are, following `def:env_realization` and `rem:exogenous_environment`, those generated by the ACS's autonomous update and the transition of the action-response component with the sequence of exogenously given background components fixed to the realized one, and those obtained by adding admissible external actions. Other sequences of background components that could have occurred are not quantified over.

Proposition (`prop:realized_reestablishment`): re-establishment on the realized trajectory. If, at some time $s > t$ on the realized trajectory, the relations of (SC-1) are judged by the rule of `def:collapse` to be established again as the same individual Σ_i, then $\mathrm{CS}_{i,t} = 0$. Likewise, for $t \in \mathbb{T}_{\mathrm{FT},i}$, if at some $s > t$ on the realized trajectory the functional activity of some basic function of (FT-1)(a), or some base function of (b), is judged by the rule of `def:functional_termination` to be established again as Σ_i, then $\mathrm{FTS}_{i,t} = 0$. *Proof sketch:* the part of the realized trajectory from t on is generated, under the realized background sequence, by Σ_i's autonomous update and the transition g_O (`rem:exogenous_environment`), so it is one of the continuations quantified in (SC-2) and (FT-2); re-establishment is judged on it, so (SC-2) (respectively (FT-2)) fails.

The converse — deriving $\mathrm{CS}_{i,t} = 1$ from the absence of re-establishment on the realized trajectory — is not stated, because the quantification in (SC-2) and (FT-2) also includes other continuations with admissible external actions added. A rescue that actually happened is described on the trajectory; admissible external actions are used only for judging recoverability.

The state indicators $\mathrm{CS}_{i,t}$ and $\mathrm{FTS}_{i,t}$ record whether each state holds at that time; the indicators $\mathrm{Col}_{i,t}$ and $\mathrm{FT}_{i,t}$ are history indicators recording that it has held at least once by time t. Given the state indicators at each time, the corresponding history indicators are fixed uniquely. These indicators and the time domain $\mathbb{T}_{\mathrm{FT},i}$ are not components of the 7-tuple, and the ACS is not required to hold or compute them internally (`subsec:description_levels_global`).

Proposition (`prop:collapse_basic`): basic properties of the functional-termination and collapse indicators. Under `def:collapse` and `def:functional_termination`:

- **(C-2) Absorption.** If $\mathrm{Col}_{i,t} = 1$, then $\mathrm{Col}_{i,s} = 1$ for all $s > t$. Likewise, if $\mathrm{FT}_{i,t} = 1$, then $\mathrm{FT}_{i,s} = 1$ for all $s > t$.
- **(C-3) Existence of a time at which the collapse state holds.** If $\mathrm{Col}_{i,t} = 1$, then (SC-1) and (SC-2) hold at some time $t' \le t$.

Proof sketch: both follow from the definitions of the history indicators. Absorption (C-2) is a property of the history indicators: $\mathrm{FT}_{i,s} = 1$ means not that Σ_i is in a functional-termination state at s, but that such a state has held by s.

Proposition (`prop:termination_order`): order of functional termination and collapse, and well-formed realization after collapse. Under `def:collapse` and `def:functional_termination`, let $\mathrm{Col}_{i,t} = 1$ and put

$$t_{\mathrm{C}} := \min\{t' \in \mathbb{N}_0 \mid \mathrm{CS}_{i,t'} = 1\}$$

(the set is nonempty since $\mathrm{Col}_{i,t} = 1$, and $t_{\mathrm{C}} \le t$). Then:

1. $t_{\mathrm{C}} \in \mathbb{T}_{\mathrm{FT},i}$ and $\mathrm{FTS}_{i,t_{\mathrm{C}}} = 1$.
2. $\mathrm{FT}_{i,t} = 1$. That is, by a time t at which collapse has occurred, a functional-termination state has already held, and the time at which a functional-termination state first holds is not later than t_{C}.
3. If $t > t_{\mathrm{C}}$, then $t \notin \mathbb{T}_{\mathrm{FT},i}$. That is, after the time at which collapse first holds, no functional-termination state is newly judged.
4. $\mathrm{WF}_{i,t} = 0$.

Proof sketch: (i) by minimality of t_{C}, $\mathrm{CS}_{i,s} = 0$ for all $s < t_{\mathrm{C}}$, so $t_{\mathrm{C}} \in \mathbb{T}_{\mathrm{FT},i}$; since $\mathrm{CS}_{i,t_{\mathrm{C}}} = 1$, the attribution provision gives that no functional activity or base function belonging to Σ_i holds at t_{C}, so (FT-1)(a)(b) hold; a continuation re-establishing any of them as the same individual would contain a realization belonging to Σ_i, which the framework does not admit from t_{C} on, so (FT-2) holds. (ii) follows from (i) and $t_{\mathrm{C}} \le t$. (iii) $s = t_{\mathrm{C}} < t$ has $\mathrm{CS}_{i,s} = 1$. (iv) follows from the attribution provision and `def:wf_realization`.

The value $\mathrm{FTS}_{i,t_{\mathrm{C}}}$ at the time t_{C} at which collapse first holds is not the value of evaluating the state of a realization after collapse; it is the judgment value at the boundary time of the termination of the same individual, fixed by (FT-1) and (FT-2) holding through the attribution provision. This boundary provision allows the case where functional termination and collapse first hold at the same time to be represented on discrete time.

Distinguishing functional termination and collapse from other states (`rem:termination_hierarchy`). The paper distinguishes:

- **Structural damage or deficiency.** A state in which some of the parts, connections, information structures or functional structures of the individual are damaged, missing, severed or degraded. As long as the organizational relations as the same individual are maintained, this is not collapse.
- **Partial malfunction or loss of function.** A state in which some of the basic functions or other functions corresponding to the primitive divisions decline or stop, or some are lost without recovery. Even if some of the basic functions corresponding to a primitive division are lost without recovery, and even if the basic functions corresponding to one or more divisions are wholly lost without recovery, as long as the activity of basic functions of other divisions or of base functions remains, this alone is neither functional termination

nor collapse. Such a loss does not cancel the identification of Σ_i as an ACS individual (the explanation following CR-2).

- **Functional termination.** The state of `def:functional_termination`: the termination of activity as an ACS individual, not the collapse of the structural individual itself.
- **Collapse.** The state of `def:collapse`: the loss, without recovery as the same individual, of the organizational relations that had constituted the identity of the structural individual. Dispersal, decomposition, dissipation, severing of connections, or disappearance of the information organization carrying structural identity can be examples, but are not the definition. Even if the constituents remain, (SC-1) and (SC-2) can hold if the organizational relations among them are lost without recovery. On the other hand, decomposition or damage after which the organizational relations can be established again as the same individual by repair or reassembly does not satisfy (SC-2).

Structural damage or deficiency can be a factor on a path to functional termination through declining function, or on a path to collapse. But the paper does not state functional termination or collapse as a general consequence of structural damage or deficiency, nor of partial loss of function, because activity as an ACS can continue after damage through redundant structure, remaining functions, alternative paths, compensation, repair or degeneration of function. Nor does the paper state collapse at the same time as a general consequence of functional termination. The only temporal relations the paper adopts are those of `prop:termination_order`: that by a time at which collapse has occurred, a functional-termination state has already held; and that after the time at which collapse first holds, no functional-termination state is newly judged.

The non-recoverability of (FT-2) is required both for the activity of the basic functions for each primitive division in (FT-1)(a) and for the base functions in (FT-1)(b). So a time at which even one basic-function activity or base function can be established again as the same individual is not a time of a functional-termination state.

The paper does not state functional termination or collapse as a general consequence of blockage $A_{i,t} = \emptyset$ (`def:blockage_sec2`), of the absence of well-formed realization $\mathrm{WF}_{i,t} = 0$, or of a time-indexed satisfaction predicate of the three principles of §3 being false. Also, $x_{i,t} \notin D_\eta$ for some $\eta \in E$ (`def:eta_satisfiable_domain`) is a state-side diagnosis that the factor cannot be satisfied because of the self-side state, and it is not identified with the stoppage of base functions in (FT-1)(b) itself.

Loss or unrecovered change of memories about oneself, modes of self-recognition or other cognitive content is distinguished from the termination of the structural self. As in `def:time_identity`, cognitive content does not affect the judgment of structural persistence, so the paper does not state functional termination or collapse as a general consequence of such losses.

Total absence of functional activity versus waiting and rest (`rem:primitive_activity_cessation`). (FT-1)(a) is a condition for excluding from functional termination the times at which part of the activity of the basic functions for the primitive divisions still holds. Sensing, discrimination, evaluation and action are abstract divisions; what is judged is not the standard maps of `def:three_primitives` themselves but the basic functions of the individual that carry the responsibilities required by CR-2. Functionally equivalent different realizations are therefore judged as activity of the same responsibility.

Selective waiting $u_\emptyset$ (`rem:action_space_structure`) is the result of selection by the functional responsibility of action, so at a time when it is selected, the functional activity of action is not absent. The same holds for pauses or rest in which sensing, discrimination or evaluation continue. Change of state by the autonomous functional process f_{auto} (`ass:f_decomposition`), and the continuation of physical change such as chemical reactions, temperature change, inertial motion or residual currents, are not by themselves activity of basic functions; so the state $x_{i,t}$ can change at a time when (FT-1)(a) holds. However, among changes represented by f_{auto}, those that are operations of base functions (`def:base_functions`) are objects of the judgment of (FT-1)(b). For example, at a time when the activity of basic functions has stopped but a base function such as transport of matter still operates, (FT-1)(b) does not hold. (FT-1) does not require the absence of all physical activity. The paper does not state (FT-1)(a) as a general consequence only of unchanged externally observed output, absence of manifest motion, or absence of change in inputs; the judgment rule for (FT-1)(a) judges whether the functional activity concerned is actually being carried.

(FT-1)(b) requires the stoppage of all base functions supporting the activity in (a), and does not prevent the constituents from physically remaining. The paper does not require (a) and (b) of (FT-1) to be mutually independent or exclusive.

Functional termination, collapse and the three principles (`rem:collapse_not_principle_failure`). (FT-1) is a condition on the activity of the basic functions for the primitive divisions and on base functions, and (SC-1) a condition on the organizational relations constituting structural identity. Both are conditions on functions or structural identity, not negations of the time-indexed satisfaction predicates of the three principles of §3. Functional termination is the state in which the activity of the basic functions for the primitive divisions and all the base functions supporting it have stopped and none recovers as the same individual; collapse is the termination of the structural individual itself. The three principles, on the other hand, are constraint axes on an existing ACS, and what has a truth value is their time-indexed satisfaction predicates (`rem:primitive_vs_principles`). Hence the paper defines neither functional termination nor collapse as a time-indexed satisfaction predicate being false, does not evaluate the time-indexed satisfaction predicates as false at times after a functional-termination state has held, and applies the judgment of the time-indexed satisfaction predicates of §3 at times with $\mathrm{FT}_{i,t} = 0$. By `prop:termination_order`(ii), $\mathrm{Col}_{i,t} = 0$ also at such times. That a quantity can still be computed formally after functional termination is distinguished from its having meaning as a judgment of a time-indexed satisfaction predicate.

Functional termination, collapse and the well-formed realization indicator (`rem:collapse_wf_relation`). $\mathrm{WF}_{i,t}$ records whether the seven components of Σ_i are given as their types prescribe in the realization of the same individual at t; it does not record whether functional activity holds. The paper therefore assumes no implication in either direction between $\mathrm{FT}_{i,t}$ and $\mathrm{WF}_{i,t}$. A time after functional termination with $\mathrm{WF}_{i,t} = 1$ — functional activity lost without recovery while the seven components are given as typed — is not excluded. To state that well-formed realization is lost after functional termination, one must show separately the antecedent of (CR-1a)(ii) of CR-1: that for some $\eta \in E$ and some $t_0 \geq 1$, $\mathrm{WF}_{i,t_0-1} = 1$ and $\eta(x_{t_0-1}, e_{t_0-1}) = 1$, and $\eta(x_s, e_s) = 0$ for all $s \geq t_0$. The paper does not state this antecedent as a general consequence of (FT-1) and (FT-2).

For collapse, by `prop:termination_order`(iv), $\mathrm{Col}_{i,t} = 1$ implies $\mathrm{WF}_{i,t} = 0$. This follows from the framework's convention (the attribution provision) that no realization belonging to Σ_i is admitted from the time of the collapse state on, together with the definition of the well-formed realization indicator; it is not a claim about the physical processes of the system. Conversely, the paper states neither $\mathrm{Col}_{i,t} = 1$ nor $\mathrm{FT}_{i,t} = 1$ as a general consequence of $\mathrm{WF}_{i,t} = 0$; a time at which well-formed realization is temporarily lost but which is neither after functional termination nor after collapse is not excluded.

Records after functional termination and after collapse (`rem:post_collapse_record`). At times after functional termination, the identification and recording of Σ_i can be maintained; the state $x_{i,t}$ and the external environment state $e_{i,t}$ can be recorded as long as the structural individual has not collapsed. However, a realization after functional termination is not treated as the realization of a currently operating ACS. After $\mathrm{Col}_{i,t} = 1$, the framework admits no realization belonging to Σ_i as the same individual, and the structural individual to which functional activity would be attributed has itself terminated. Times after collapse are not treated as times at which "a functional-termination state continues" (`prop:termination_order`(iii)). CR-2 is not a diagnostic value at each time but a condition identifying Σ_i as an ACS individual (`rem:math_vs_realization`); the paper does not re-judge, time by time, a realization after functional termination or after collapse as a currently operating ACS satisfying CR-2 anew. Nor are the earlier identification of Σ_i as an ACS individual and its individual identification cancelled retroactively by partial loss of function, total loss of function of one or more divisions, functional termination or collapse.

The ACS-record-valid time domain of §7 consists of the times with $\mathrm{Col}_{i,t} = 0$ and can include times after functional termination. Records there are records of tracking as a structural individual; the description of functions as a currently operating ACS is restricted to times with $\mathrm{FT}_{i,t} = 0$. How state bands and paths of reaching are described for the two stages, functional termination and collapse, is treated in §7.